

Mikko Numminen

AI engineering and full-stack development

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PROFILE Full-stack developer working on systems built around language models, with a focus on what happens when the model is wrong. I keep deterministic work deterministic, put the model only where language genuinely cannot be rule-coded, and enforce grounding through validation rather than prompt wording. I measure results and publish corrections when a measurement turns out to be broken. Moved into software in 2022 after more than twenty years running retail and hardware store operations.

STACK	Languages	TypeScript, JavaScript, Python, C#, Rust, SQL, HTML/CSS, Bash, PowerShell
	AI and LLM	Agent and pipeline design, MCP, evaluation harness design, blind A/B model comparison, grounding validation and citation checking, structured-output salvage and schema validation, prompt-injection containment and red-team evaluation, cost and latency measurement, RAG with hybrid retrieval and relevance gating, pgvector, local serving with Ollama (Poro 2, Qwen3, Llama 3.1), Microsoft.Extensions.AI provider abstraction, Claude Code multi-agent orchestration, TTS engines
	Backend	REST API design, PostgreSQL, .NET 8 and ASP.NET Core, EF Core, ASP.NET Core Identity, Node.js, FastAPI, Prisma, Kysely, NextAuth v5, 2FA and TOTP, RBAC and audit trails, SQLite, MongoDB, Neon
	Frontend	React 19, TypeScript strict mode, Next.js 15/16, Astro, Tailwind CSS v4, MUI X Charts, Three.js, GSAP, Phaser 4, WebAssembly (Rust to wasm32 via wasm-bindgen), CSP and security headers
	Data	Data visualisation for decision-making, large-scale open-data processing, schema design and migrations, deterministic seeded data generation, synthetic corpora as a GDPR decision
	Platform	Docker and Compose, GitHub Actions, CI gating with branch protection, Azure (App Service, Static Web Apps), Vercel, Kubernetes and Helm, Turborepo, Cloudflare Tunnel and Access, Tailscale, self-hosted deployment
	Testing	Evaluation as a blocking CI job, Vitest, Playwright E2E, coverage thresholds enforced by pre-push hooks, known-answer test vectors, tamper-detection tests, snapshot fallback modes, Sentry, OpenTelemetry
	Security	Zero-knowledge architecture, threat modelling, Argon2id, XChaCha20-Poly1305, AEAD with associated data binding, zeroize and secrecy, constant-time comparison, secret-hygiene auditing
	Practice	ADR-driven design, numbered decision records, published methodology and public corrections, AI-assisted development as daily practice

PROJECTS **Feedback Intelligence** C# / .NET 8 / local LLM · mikkonumminen.dev/feedback

Domain-agnostic engine that turns messy free-text feedback into a grounded situational view. A deterministic alert layer runs before the model; theme counts and trend direction are computed arithmetic. Synthesis must cite the feedback items it drew on, and a narrative whose citations fail validation is dropped to a deterministic fallback with the drop logged, so an untraceable claim cannot reach the view. Structured output passes through a mandatory salvage ladder ending in a preserved-raw failure record. Model chosen by blind comparison rather than reputation. Decisions recorded as numbered ADRs.

Finnish LLM evaluation published methodology

Blind round-by-round comparison of Finnish language quality across three local models. Retracted and corrected my own published containment finding after tracing it to too few rounds and a rate limiter corrupting the measurement; clean re-measurement showed no real difference and the incorrect claim was removed from the live site. The harness now runs as a blocking CI job under branch protection.

PasswordManager

Rust workspace

Local-first zero-knowledge password manager. One Rust crypto core shared by CLI, sync server and browser, with no second implementation to drift. The same core compiles to wasm32 and is driven from a small hand-written JS client, so decryption happens in the browser and the server stays zero-knowledge on the web path too. Argon2id and XChaCha20-Poly1305 from vetted crates, ciphertext bound to entry UUID and timestamp through associated data. Tests include known-answer vectors, tamper detection, a check that inspects the raw server database for plaintext after real pushes, and two-device sync over real HTTP.

Portfolio RAG

FastAPI / pgvector / Ollama

Self-hosted retrieval system behind a terminal-style contact interface. Hybrid retrieval, relevance gating so out-of-scope questions are refused rather than improvised, prompt-injection containment, in-process embeddings, no hosted model anywhere in the path.

HRManager

Next.js 16 / React 19 / Prisma

Production web app with 2,910 tests at 92.2% line coverage across UI component, server-action and Playwright E2E suites. RBAC, dashboard analytics, immutable audit trail.

Platform (vuohiliitto.com)

Next.js 15 / Turborepo

Multi-tenant community platform in production. 1,388 tests across 153 suites plus 42 Playwright E2E with a coverage threshold enforced by pre-push hook. Gamification system, Blizzard API and Raider.IO integration.

AudiobookMaker, ReadLog, Spacepotatis

Python, C#, TypeScript

Shipped desktop TTS application with pluggable engines, a Finnish text normalizer and 2,172 tests; merged an upstream bugfix into resemble-ai/chatterbox. ReadLog ported from Next.js and Prisma to ASP.NET Core with EF Core, deployed to Azure via GitHub Actions. Spacepotatis is a Phaser 4 browser game with roughly 1,170 tests.

EXPERIENCE

Kasvu Labs Oy, Software Developer

2022 – 2024

Full-stack development on Node.js and React with Azure. Built data-visualisation tooling to make large datasets graspable and turn them into decisions, alongside large-scale open-data processing, database management, UI design and product maintenance.

Keijo Numminen Oy, Kesko Oyj, K-Niemi Oy

1998 – 2022

Senior salesperson and store operations. More than twenty years in K-Rauta and retail: supervising and training staff, directing daily work, procurement, inventory management and logistics for private and professional customers.

EDUCATION

Tampere University

in progress

Computer science, studies in progress.

LANGUAGES

Finnish (native) · English (professional)